Pointwise covariance determines the projective Dunford moment

An affirmative answer to Problem 9.5 of arXiv:2004.00237

Candidate solution packet

August 11, 2026

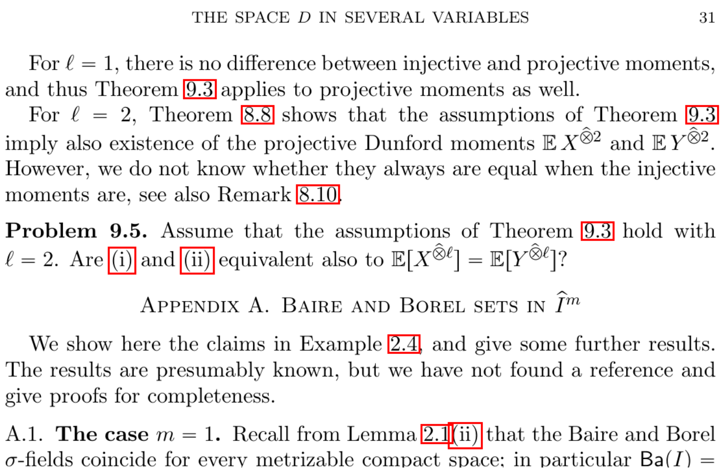
Status. Candidate full solution, likely valid pending expert review.

1 Source question

Janson identifies $D([0,1]^m)$ isometrically with $C(K)$, where $K = \widehat{I}^m$ and $\widehat{I}$ is the split interval. The source paper's Theorem 9.3 says that, for D-measurable $D([0,1]^m)$-valued random variables X, Y satisfying

$$M_X := \sup_{t\in[0,1]^m} \mathbb{E}|X(t)|^2 < \infty, \qquad M_Y := \sup_{t\in[0,1]^m} \mathbb{E}|Y(t)|^2 < \infty,$$

equality of the pointwise second moments at all pairs of split points is equivalent to equality of the injective Dunford second moments. Theorem 8.8 also gives existence of the corresponding projective Dunford moments. Problem 9.5 asks whether those projective moments must be equal [1].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For $\ell = 1$, there is no difference between injective and projective moments, and thus Theorem 9.3 applies to projective moments as well.

For $\ell = 2$, Theorem 8.8 shows that the assumptions of Theorem 9.3 imply also existence of the projective Dunford moments $\mathbb{E}\, X^{\widehat{\otimes}2}$ and $\mathbb{E}\, Y^{\widehat{\otimes}2}$. However, we do not know whether they always are equal when the injective moments are, see also Remark 8.10.

Problem 9.5. Assume that the assumptions of Theorem 9.3 hold with $\ell = 2$. Are (i) and (ii) equivalent also to $\mathbb{E}[X^{\widehat{\otimes}\ell}] = \mathbb{E}[Y^{\widehat{\otimes}\ell}]$?

Appendix A. Baire and Borel sets in $\widehat{I}^m$

We show here the claims in Example 2.4, and give some further results. The results are presumably known, but we have not found a reference and give proofs for completeness.

A.1. **The case** $m = 1$**.** Recall from Lemma 2.1(ii) that the Baire and Borel σ-fields coincide for every metrizable compact space; in particular $\mathsf{Ba}(I) =$

Figure 1: Problem 9.5 on source PDF page 31.

Equivalently, the question is whether

$$\mathbb{E}[X(s)X(t)] = \mathbb{E}[Y(s)Y(t)] \qquad (s,t\in K) \tag{1}$$

forces

$$\mathbb{E}\gamma(X,X) = \mathbb{E}\gamma(Y,Y) \tag{2}$$

for every bounded bilinear form γ on $C(K)$.

2 Proof intuition

Grothendieck's theorem extends any fixed bilinear form on $C(K)$ to a Hilbert space $H = L^2(K,\nu)$. The apparent obstruction is that a D-measurable random function need not be jointly measurable on $\Omega \times K$, so it is not immediate that it is a genuine random variable in H.

The key observation is measure-theoretic rather than topological. For every finite Baire measure on K, its completed measure algebra is countably generated. Indeed, in each split-interval coordinate, the only information beyond the usual Borel structure of $[0,1]$ is a countable correction; modulo a fixed measure, only the countably many positive singleton atoms matter. Thus $L^2(K,\nu)$ is separable.

The special Baire-Fubini theorem proved in the source paper then shows that all scalar H-coordinates of X are measurable and that X has finite Hilbert second moment. Separability upgrades weak measurability to strong measurability. Applying the same Fubini theorem to the random function $(s,t) \mapsto X(s)X(t)$ turns (1) into equality of all matrix entries of the Hilbert covariance operators. Those are positive trace-class operators, hence the Hilbert projective second moments agree. The Grothendieck extension then yields (2).

3 A separability lemma

We write $\rho : \widehat{I} \to I = [0,1]$ for the map identifying the two copies of each split point. Proposition A.1 of [1] states that

$$\mathrm{Ba}(\widehat{I}) = \{\rho^{-1}(B) \bigtriangleup N : B \in \mathcal{B}(I),\ N \subseteq \widehat{I} \text{ countable}\}. \tag{3}$$

The same paper recalls that Baire sigma-fields commute with finite products.

Lemma 1. *Let $r \geq 1$, let $K_r = \widehat{I}^r$, and let ν be a finite positive Baire measure on K_r. Then $L^p(K_r,\nu)$ is separable for every $1 \leq p < \infty$.*

Proof. Put $\Sigma_0 = \mathrm{Ba}(\widehat{I})$, so that $\mathrm{Ba}(K_r) = \Sigma_0^{\otimes r}$. Let ν_j be the j-th marginal of ν, and let

$$A_j := \{x \in \widehat{I} : \nu_j(\{x\}) > 0\}.$$

The set A_j is countable. Let $\mathcal{G}_j$ be the sigma-field generated by all $\rho^{-1}(B)$, with B ranging over a fixed countable generator of $\mathcal{B}(I)$, together with the singletons $\{x\}$ for $x \in A_j$. Thus $\mathcal{G}_j$ is countably generated.

If $E \in \Sigma_0$, write $E = \rho^{-1}(B) \triangle N$ as in (3). The countable set $N \setminus A_j$ is ν_j-null, while $N \cap A_j \in \mathcal{G}_j$. Hence E agrees ν_j-almost everywhere with a member of $\mathcal{G}_j$.

Set $\mathcal{G} = \mathcal{G}_1 \otimes \cdots \otimes \mathcal{G}_r$, again a countably generated sigma-field. Every coordinate cylinder from Σ_0 agrees ν-almost everywhere with a cylinder from the corresponding $\mathcal{G}_j$, because the inverse image of a ν_j-null set under the coordinate projection is ν-null. The class of sets in $\Sigma_0^{\otimes r}$ which agree ν-almost everywhere with a member of $\mathcal{G}$ is a sigma-field. It contains all coordinate cylinders, and therefore equals $\Sigma_0^{\otimes r}$.

Thus the completion of $\mathrm{Ba}(K_r)$ modulo ν is already the completion of a countably generated sigma-field. Simple functions from a countable generating algebra are dense in $L^p(K_r,\nu)$, proving separability. □

4 Main result

Theorem 1. *Let $m \geq 1$, $K = \widehat{I}^m$, and $E = D([0,1]^m) = C(K)$. Let X, Y be D-measurable E-valued random variables satisfying $M_X, M_Y < \infty$. If (1) holds for all $s,t \in K$, then (2) holds for every bounded bilinear form $\gamma : E \times E \to \mathbb{R}$. Consequently*

$$\mathbb{E}[X \widehat{\otimes}_\pi X] = \mathbb{E}[Y \widehat{\otimes}_\pi Y] \quad \textit{in } (E \widehat{\otimes}_\pi E)^{**}.$$

In particular, the answer to Problem 9.5 of [1] is affirmative.

Proof. Fix a bounded bilinear form γ on $E = C(K)$. By the form of Grothendieck's theorem used in the proof of [1, Theorem 8.8], there are a Baire probability measure ν on K and a bounded bilinear form $\widetilde{\gamma}$ on

$$H := L^2(K, \nu)$$

whose restriction to $C(K) \times C(K)$ is γ.

We first justify treating X and Y as square-integrable, strongly measurable H-valued random variables. Let $h \in H$, choose a Baire representative, and define the finite signed Baire measure $\lambda_h(A) = \int_A h\, d\nu$. Since $\sup_t \mathbb{E}|X(t)| \le M_X^{1/2}$, the Baire-Fubini theorem [1, Theorem 6.1(iii)], applied to X and λ_h, shows that

$$\omega \longmapsto \int_K X(\omega, s) h(s)\, d\nu(s) = \langle X(\omega), h\rangle_H$$

is measurable. Thus $X : \Omega \to H$ is weakly measurable. Lemma 1 makes H separable, so Pettis' measurability theorem makes X strongly measurable. The same holds for Y.

Apply the positive part of the source's Baire-Fubini theorem to the $D([0,1]^m)$-valued random variable $|X|^2$ and the measure ν. The source's extension bound from ordinary to split points gives

$$\mathbb{E}\|X\|_H^2 = \int_K \mathbb{E}|X(s)|^2\, d\nu(s) \le M_X.$$

Likewise $\mathbb{E}\|Y\|_H^2 \le M_Y$. Therefore the Bochner projective tensors

$$T_X := \mathbb{E}[X \widehat{\otimes}_\pi X], \qquad T_Y := \mathbb{E}[Y \widehat{\otimes}_\pi Y] \quad \text{in } H \widehat{\otimes}_\pi H$$

exist, since $\|x \otimes x\|_\pi = \|x\|_H^2$.

We next compare their matrix entries. For $h, k \in H$, let $\lambda_h = h\nu$ and $\lambda_k = k\nu$ as above. The function

$$Z_X(\omega; s, t) := X(\omega, s) X(\omega, t)$$

is a $D([0,1]^{2m}) = C(K \times K)$-valued random variable: every sample is continuous on $K \times K$, and each ordinary point evaluation is a product of two measurable evaluations of X. Moreover, Cauchy–Schwarz and the split-point extension bound give

$$\sup_{s,t \in K} \mathbb{E}|Z_X(s,t)| \le M_X.$$

Theorem 6.1(iii) of [1], now on $K \times K$ with the finite signed Baire measure $\lambda_h \otimes \lambda_k$, yields

$$\mathbb{E}[\langle X, h\rangle_H \langle X, k\rangle_H] = \int_{K \times K} \mathbb{E}[X(s)X(t)]\, h(s) k(t)\, d\nu(s) d\nu(t).$$

The same formula holds for Y. Assumption (1) therefore implies

$$\mathbb{E}[\langle X, h\rangle_H \langle X, k\rangle_H] = \mathbb{E}[\langle Y, h\rangle_H \langle Y, k\rangle_H] \qquad (h, k \in H). \tag{4}$$

Under the standard isometric identification of $H \widehat{\otimes}_\pi H$ with the trace-class operators on the real Hilbert space H, T_X and T_Y are the positive covariance operators whose matrix entries are the two sides of (4). Hence (4) implies $T_X = T_Y$. Finally, $\widetilde{\gamma}$ is a continuous functional on $H \widehat{\otimes}_\pi H$, so

$$\mathbb{E}\gamma(X, X) = \langle \widetilde{\gamma}, T_X\rangle = \langle \widetilde{\gamma}, T_Y\rangle = \mathbb{E}\gamma(Y, Y).$$

Since γ was arbitrary, the projective Dunford moments in $(E \widehat{\otimes}_\pi E)^{**}$ are equal. Their existence under the stated hypotheses is also Theorem 8.8 of [1]. ☐

5 Consequences and limitations

Combining Theorem 1 with Theorem 9.3 of [1], all three conditions are equivalent under the uniform pointwise second-moment bounds: equality of split-point covariances, equality of injective Dunford second moments, and equality of projective Dunford second moments.

The argument is specific to the second moments occurring in Problem 9.5. It does not prove that the canonical map $(E\widehat{\otimes}_\pi E)^{**} \to (E\widehat{\otimes}_\varepsilon E)^{**}$ is injective on all of the bidual, the broader question in Remark 8.10. It instead proves injectivity on the particular class of differences of Dunford covariance moments satisfying the source's pointwise L^2 bounds.

6 Verification and novelty status

The proof was checked specifically for four common hidden gaps: the nonmetrizable Baire measure is reduced only modulo the actual Grothendieck measure; weak measurability is upgraded only after L^2-separability is proved; every double integration uses the source's special Fubini theorem on $D([0,1]^{2m})$; and equality of elementary matrix entries is lifted through the faithful trace-class representation of the Hilbert projective tensor.

On August 11, 2026, the local run indexes were searched by arXiv id, title, and core terms. Targeted web searches for the exact question and close phrases found the source paper and the earlier background monograph [2], but no later resolution. The search was bounded, so novelty confidence is moderate pending a specialist citation review.

References

[1] S. Janson, *The space D in several variables: random variables and higher moments*, Math. Scand. **127** (2021), no. 3, 544–584; arXiv:2004.00237.

[2] S. Janson and S. Kaijser, *Higher moments of Banach space valued random variables*, Mem. Amer. Math. Soc. **238** (2015), no. 1127; arXiv:1208.4272.

[3] A. Grothendieck, *Résumé de la théorie métrique des produits tensoriels topologiques*, Bol. Soc. Mat. São Paulo **8** (1953), 1–79.